

Accelerating Knowledge Transfer by Sensing and Actuating Social-Cognitive States

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ABSTRACT

Knowledge Transfer is one of the essential principles in education. The teacher's knowledge is encoded through speech and writing and transmitted to the student. The student then decodes the transmitted information according to individual capabilities and absorbs it as knowledge. This paper presents an approach to accelerate knowledge transfer using sensor technology and social-cognitive states. So far, we have worked on quantifying meeting discussions, analyzing lecture studies, and estimating domain knowledge from web browsing. The contribution of this study is to estimate the degree of achievement of knowledge transfer and to accelerate it based on the estimated results.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Education**.

KEYWORDS

Education, e-Learning, Learning Analytics, Knowledge Transfer

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PROBLEM STATEMENT

In the concept of learning lifecycle, *Perceive*, *Master*, and *Transfer* are important factors [32]. *Perceive* is a process of gaining knowledge from a textbook or lecture. *Master* is a process of perceiving knowledge usability by conducting exercises and experiments. *Transfer* is a process of transferring mastered knowledge to one another by conducting a presentation or discussion. In the field of ubiquitous computing, Ishimaru and Dengel worked mainly on the analysis and acceleration of *Perceive* and *Master* [11]. However, analysis and acceleration of *Transfer* is yet to discover.

Understanding and accelerating knowledge transfer is essential in all kinds of fields. Allen et al. investigates that 11 million workplace meetings are held every day in the United States [1]. Employees spend an average of six hours in scheduled weekly meetings, with supervisors spending 23 hours [23]. Meetings are time-consuming for attendees and costly to the organization, ranging from 30 million to over 100 million US dollars annually [24]. Due to the importance of the field of study on knowledge transfer, the number of related publications is increasing every year [9]. Research on knowledge transfer is gaining attention and importance every year.

According to Shanon-Weaver Model [26], *Knowledge Transfer* is defined as communication between the sender (transmitter) and the receiver using encoding and decoding knowledge. Previous studies work on the implementation of a sensor environment for estimation of learning difficulty [30, 40], estimation of wakefulness while taking a video lecture [12], and behavioral or cognitive-state estimation in the online meetings [35, 36]. Our study aims to quantify and accelerate the success of encoding and decoding knowledge.

This paper discusses the current findings and planned schedule for knowledge transfer. The specific problems addressed in this work are three-fold. (1) How to estimate the state of knowledge transfer? (2) How to visualize the state of knowledge transfer? (3) How to accelerate the knowledge transfer?

RELATED WORK

Several studies focus on the analysis of multi-person sensing related to knowledge transfer. Here we highlight works related to the factor of knowledge transfer, estimation of knowledge transfer, and acceleration of knowledge transfer.

Factor of Knowledge Transfer

To understand knowledge transfer, estimating or recognizing the activities and behaviors that occur in communication is essential. In communication, there are three key features, *Participants Appearance*, *Verbal Communication*, and *Nonverbal Communication* [7]. *Participant Appearance* represents features concerning the appearance of the participants in the communication. Maria Schulte et al. found that in the communication scene, participant age and distribution have a significant positive relationship with forgiveness and the number of counteractive behaviors [17]. Due to the importance of participants' age for information transfer, Geng et al. has worked on age estimation from facial images using open source datasets such as the FG-NET Aging Database [15] and the MORPH Database [22]. *Verbal Communication* represents characteristics related



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Figure 1: A figure shows an experimental target of the study on analyzing and accelerating knowledge transfer in education.

to speech. Related research focuses on the analysis of conversational content [19], pronunciation [27, 41], speech volume [13], and voice pitch [18]. *Nonverbal Communication* refers to behavior-related characteristics that do not include speech. The related study focuses on the analysis of posture [4, 25], smile [5, 43], eye contact [38, 42], and head motion [21, 39]. Previous studies work on social activity behavior analysis in communication. However, some factors of knowledge transfer, such as micro-behavioral analysis and engagement detection, are still being studied.

Estimation of Knowledge Transfer

Estimating the state of knowledge transfer is presented by Srivastava [28]. The study works on the estimation of learning difficulty using contactless sensors. The study presents the use of the slider to collect a real-time understanding of the labeling. The contactless sensors are a Logitech webcam, a Tobii Pro X2-30 eye-tracker, and an Optris PI-400 thermal camera. After collecting data from 100 participants, the study found that the difficulty of the lecture depended on three factors: the pictorial representation of complex terms, the difference between the amount of verbal and visual information presented per slide, and the number of words spoken by the instructor per second [30]. Babaei et al. is then working on automatic tracking of attentional states using a webcam [2]. As a result, action units of the face, such as lips or eyelids, perform well in attention detection. Another work was done to discover the correlation between attention while watching videos and comprehension level [29]. Participants with higher prior knowledge made fewer eye movement transitions than those with lower prior knowledge. Prior knowledge estimation has been done by analyzing the web browsing logs [20]. Hence the direction of web browsing analysis can perform well on knowledge transfer estimation. While the above studies deal with comprehension estimation, they focus on something other than the breakdown of comprehension into details of encoding and decoding in knowledge transfer.

Acceleration of Knowledge Transfer

Knowledge transfer is accelerated based on the state of knowledge transfer estimation results. Kuttal et al. work on human-human and human-agent comparison on pair programming [14]. The study discovered no significant difference in productivity, self-efficacy, code quality, and pair programming preferences when using an agent for pair programming. Test-driven programming can be done together with an agent. The acceleration of knowledge transfer happens towards using the agent can provide code skeletons and hints to lead their partner toward a solution. Haliburton et al. use wristband devices to give thermal feedback to the presenter to

feedback audience engagement [10]. The work could enhance socio-emotional connectedness in virtual and hybrid meetings. As a result, the presenter and audience feel connectivity only if they are in the same atmosphere, which enhances the presenter's confidence in transferring knowledge. Some study works on the acceleration of knowledge transfer. However, there still needs to be more work on enhancing communication.

METHODOLOGY

To answer three research questions highlighted in the Problem Statement, we mention methods such as experimental target conditions, data acquisition sensors, and data analysis.

Experimental Target Condition

Figure 1 presents the experimental target scenes for the analysis of knowledge transfer. Since COVID-19, we have tackled the difficulty in the form of changes in knowledge transfer occurring in the educational field. The knowledge transfer occurs in onsite, online, or hybrid form. In order to cover the condition of research in the wild condition, we work on the analysis of knowledge transfer in hybrid conditions. In this study, we aim to discover knowledge transfer occurs in onsite/online meetings, onsite/online lectures, and web browsing.

Data Acquisition Sensors

For a variety of conditions, our key sensors used are followed. In order to allow using our proposed application or system, we focus on using minimum sensors. Especially with work from the home era, built-in camera roles are significant. Therefore, we use a webcam as an option for sensing. Another sensor is to use an eye-tracker. The analysis of gaze roles is significant in understanding participants' cognitive states. Hence, we use eye-tracker in studies to further understand participant cognitive behavior. Lastly, we use web applications for annotation purposes or data collection. Especially, TrackThinkTS [16] is used for the web browsing log collection.

Data Analysis

Our study focuses on using the camera as a sensor. Due to the choice of sensor, we come up with using OpenFace [3]. Also, our target is knowledge transfer-related behavior analysis, so we use the sliding window technique. This approach can vectorize the OpenFace data, and majority voting will improve accuracy in prediction. The prediction model will also use the standard approach to understand important features. After the feature extraction approaches, deep learning approaches are applied. By applying several deep learning

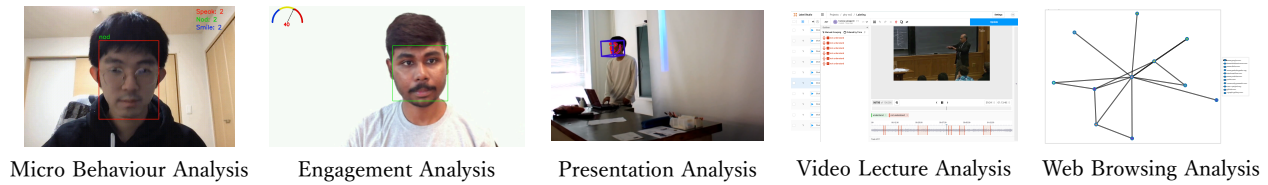


Figure 2: A figure shows an already published result and current on going analysis.

models, performance concerns like real-time processing will be the main factor. In the end, apply the deep learning model to implement a software application.

RESEARCH CONDUCTED AND FUTURE PLAN

Figure 2 shows examples of research conducted and planned. This section explains the progress in each meeting analysis, lecture analysis, and web browsing analysis section.

Onsite and Online Meeting Analysis

Meeting analysis is essential for knowledge transfer analysis. During the meeting, participants can be both knowledge sender and receiver. We will then work on the recognition of behavior that roles are significant for both roles.

The *Discaas* project [36] aims to recognize micro behavior occurring in multi-person meetings. Currently, the dataset contains 295 videos (total video length 21.7 hours) from 40 participants, of which 190 videos of 13.2 hours are labeled. For three detection models, the data contains 4.19 hours of nod data, 12.9 hours of speaking data, and 7.92 hours of smile data. Note that all videos are at 25 frames per second (fps). Micro-behavior mainly targets nodding. Also, we challenge recognizing speaking actions from mouth movement due to privacy concerns. The recognition uses OpenFace and a sliding window approach to vector the features. As a result, *speaking* and *nodding* were recognized with an average macro f1-score of 67.9% in a 10-fold random split cross-validation and an average macro f1-score of 62.5% in a leave-one-participant-out cross-validation. In order to increase the accuracy of the model, the deep learning approach is applied together with Chen et al. [6]. The neural-network-based end-to-end system is called as MeetingPipe. For head rotation, data was acquired as the primary nod feature for nod detection, and then temporal information on the nod pattern was acquired from the head pitch angle using a recurrent gate unit and a squeeze and excite mechanism. For speaking recognition, we use TalkNet, an active speaker detection model that can effectively recognize active speakers from videos. Lastly, smile detection uses a neural network framework consisting of an 18-layer residual network for feature representation and a self-attention layer to search for correlations in each receptive field. As a result, the F1 scores of the *nodding*, *speaking*, and *smiling* detection are 81.26%, 94.90%, and 97.34%, respectively.

The *EnGauge* project [35] aims to detect the engagement level of participants in online meetings. The data includes video recordings of 24 people with low, medium, and high levels of engagement. The data is collected using a role-acting approach, in which each participant is asked to act out a different engagement. As in the

previous study, we first applied OpenFace and the sliding window approach, and the F1 score of the classification was 46.7%. To improve the classification rate, MobileNetV2 is applied, and the F1 score is 89.5% for leave-one-participant-out cross-validation. Using the MobileNetV2 model, we implement a gauge-like user interface application. In the current EnGauge application, the engagement recognition is done by frame image, and the result will be present as a gauge interface as shown in Figure 2.

The *DiscussionJockey* project [31] uses background music as an interrupter in online meetings. The study aims to allow online meeting participants to speak equally. To do so, we dynamically change the music speed to change the participants' utterance volume naturally. As a result, dynamic change in background music had some potential to balance the amount of speech.

The planned study uses these recognition models to implement a dashboard for visualizing each status during meetings. Also, accelerate the knowledge transfer by presenting real-time feedback on the engagement in the meetings.

Onsite and Online Lecture Analysis

Lecture analysis is another important field of education research. In the current progress, we are working on collecting some lecture data. We have conducted a study on collecting onsite lectures. We collect the presenter's body using a camera in the data collection. On behalf of collecting audiences using another camera. In order to evaluate the level of knowledge transfer, we collect subjective feedback on the score against the presentation. From the dataset, we apply OpenFace to see the correlation between the score of the subjective feedback and the presenter's behavior. Once we discover important features, as the other projects were done, we are eager to apply deep learning techniques to estimate the successful presentation or knowledge transfer. And not only the presenter's analysis but the audience's behavior is a scope of analysis as well.

The online lecture data collection was done using video content. We collect data from participants watching video lectures. Inspired by Srivastava [28], we use an eye-tracker and webcam to collect how participants take the online lecture. In the previous research, the study did not collect comprehension questions regarding each time segment. Meaning that real-time self-annotated comprehension level is collected but does not check the actual comprehension on each time segment. Therefore, we decide to prepare questions regarding each time segment to collect self-reported and actual participant understandings of the online lecture. We will discover a correlation between comprehension levels using sensor data collected by the webcam and the eye-tracker.

In onsite and online lecture analysis, we aim to estimate the comprehension level of audiences. Using the comprehension estimation model, we will implement a dashboard to visualize the success and failure of knowledge transfer for both students and teachers. Also, accelerate the knowledge transfer by giving real-time feedback on the audience's comprehension to the presenter.

Web Browsing Analysis

Web browsing analysis is essential for the target audience to understand new and existing knowledge. Personalization is one of the key concepts in the field of knowledge transfer. Hence, understanding what participants know or experience is crucial to discover patterns of decoding [37]. The level of understanding may vary depending on the person's domain knowledge. Nagano et al. has proposed an idea of visualizing the way of thinking by collecting web browsing logs [20]. We follow the concept of learning to develop Google Chrome Extension [16] further to collect web browsing logs. Using TrackThinkTS, we collect two groups of programmers, 20 novices and 13 experts [34]. Experts refer to the participants who took university lectures and are experienced in the Scheme language. The study aims to discover domain knowledge using the web search logs and code compile logs collected by C2Room. As a result, the binary classification of novice and experienced participants using Random Forest achieved an average precision rate of 0.95. We then implement the knowledge graph from the exported data of TrackThinkTS [33]. The work supports us in visualizing participants' existing knowledge. The findings will be used to match knowledge between the sender and receivers. In the future, automation of knowledge encoding will be done using TrackThinkTS and then deliver the knowledge to the necessary participants.

DISCUSSION

In this section, we discuss three research questions stated in Problem Statement.

How to estimate the state of knowledge transfer?

The state of knowledge transfer is estimated using sensors like cameras, eye-tracker, and web browsing loggers. We collect participants' data in onsite/online meetings, onsite/online lectures, and during web browsing. We estimate the state of knowledge transfer mainly from the self-report and question-solving tasks. Knowledge transfer estimation will be successful using sensor data and ground truth comprehension labels.

How to visualize the state of knowledge transfer?

The visualization of the state of knowledge transfer will be done by implementing a software application. Currently, visualization for micro-behavior [36] and engagement [35] recognition is implemented. The next step is to visualize all the related factors of knowledge transfer estimation into a dashboard. Real-time visualization is also in the scope of the study.

How to accelerate the knowledge transfer?

The visualization application we have developed accelerates knowledge transfer. We will identify any changes in the efficiency of knowledge transfer before and after using this application. The

results will be evaluated through statistical testing of existing applications.

CONCLUSION

This paper presents approaches for estimating, visualization and accelerating knowledge transfer. The estimation has been done on micro-behaviors, engagement, and comprehension. The visualization is done on micro-behaviors and engagement. Lastly, acceleration is ongoing work. We will further work on the research plan.

CONTRIBUTION IN THE FIELD OF UBICOMP

The expected contributions of this research are to investigate sensing approaches to estimate, visualize, and accelerate knowledge transfer. The study will be significant for the field of ubiquitous computing to the educational field to enhance both student and teacher learning. In the long term, the core application will be used in ubiquitous computing in educational and wild-like office environments.

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